Progress Note: 2-Period Igami Model with Agglomeration

14.273 Mini Project

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1 Overview

This note documents the first extension of Igami (2017) to incorporate **innovation spillovers / agglomeration effects** in a simplified 2-period setting. The goal is to understand whether these effects undo the innovator's dilemma found by Igami.

2 Model

2.1 State and Firm Types

The market state at each period t is $s_t = (n_o, n_b, n_n, n_{pe})$, where:

- n_o: old incumbents (have *not* adopted new technology),
- n_b: "both" incumbents (have adopted; produce with new tech),
- n_n: new entrants (using new technology),
- n_{pe}: potential entrants.

Active firms in the product market are those with $n_o + n_b + n_n > 0$.

"Both" firms are multi-product: they continue to sell old-gen (at cost c_o) while also selling new-gen (at cost $c_{n,t}$). This is the source of the **cannibalization effect**: by expanding new-gen supply, a "both" firm depresses the old-gen price it receives, creating a disincentive to innovate that is absent for pure new entrants.

2.2 Timing Within Each Period

1. **Product market:** Cournot competition in *two differentiated markets* (old-generation and new-generation products) with linear inverse demand:

$$P_o = A - \frac{B}{M} Q_o - \frac{\rho B}{M} Q_n, \tag{1}$$

$$P_n = A - \frac{B}{M} Q_n - \frac{\rho B}{M} Q_o, \tag{2}$$

   where $\rho \in [0, 1)$ is the cross-market substitution parameter. Old firms (n_o) compete only in the old-gen market at cost c_o. New entrants (n_n) compete only in the new-gen market at cost $c_{n,t}$. "Both" firms (n_b) are *multi-product*: they sell in both markets simultaneously and internalize the **cannibalization** effect, in that their new-gen output reduces demand for their own old-gen product.

2. **Private shocks:** Each firm draws i.i.d. cost shocks $\varepsilon_{i,a} \sim \text{EVT1}(0, \sigma)$ for each action a.

3. **Discrete choices** (sequential order: old $\to$ both $\to$ new $\to$ pe):

   - old: $\{\text{stay}, \text{innovate}, \text{exit}\}$
   - both/new: $\{\text{stay}, \text{exit}\}$
   - pe: $\{\text{enter}, \text{quit}\}$

   Each type observes the *realized* outcomes of all earlier movers before making its own decision.

4. **Transition** to period $t + 1$.

2.3 Agglomeration Extension

The new-technology marginal cost depends on the number of firms already using it:

$$c_{n,t} = c_{n,0} - \gamma \cdot (n_{b,t} + n_{n,t}), \qquad \gamma \geq 0. \tag{3}$$

When $\gamma = 0$ the model reduces to the Igami baseline. When $\gamma > 0$, a larger innovative cluster lowers costs for all new-tech firms, capturing the knowledge-spillover or agglomeration channel.

2.4 Value Functions and CCPs

Because firms move sequentially within each period and each type observes the *realized* outcomes of all earlier movers, each type's decision problem is effectively a *single-agent problem*: earlier-mover uncertainty has been resolved, and later-mover choices enter only

through pre-computed CCPs. Combined with i.i.d. EVT1 private cost shocks, this yields closed-form logit CCPs at each stage:

$$P(a \mid \tau, \tilde{s}) = \frac{\exp\big(\bar{V}_a(\tau, \tilde{s})/\sigma\big)}{\sum_{a'} \exp\big(\bar{V}_{a'}(\tau, \tilde{s})/\sigma\big)}, \tag{4}$$

where $\tilde{s}$ is the *realized intermediate state* after all earlier movers have acted, and $\bar{V}_a$ is the systematic utility:

$$\bar{V}_a(\tau, \tilde{s}) = \pi_\tau(s_0) + \text{cost}(a) + \beta\, \mathbb{E}[V_{t+1}(\tau', s') \mid \tilde{s},\, a_i = a,\, \text{CCP}_{\text{later}}]\,. \tag{5}$$

Here $\text{CCP}_{\text{later}}$ denotes pre-computed CCPs of same-type peers and all later-moving types; flow profits $\pi_\tau(s_0)$ are evaluated at the period-opening state.

2.5 Solution Method

The model is solved by backward induction, following Igami (2017):

1. **Terminal period ($t = 1$):** $V_1(\tau, s_1) = \pi_\tau(s_1)$, where Cournot profits use the agglomeration-adjusted cost $c_{n,1} = c_{n,0} - \gamma(n_{b,1} + n_{n,1})$.

2. **Period $t = 0$:** For each state s_0, stages are solved in *reverse move order* (PE $\to$ new $\to$ both $\to$ old), exploiting the sequential structure:

   - **Stage 4 (PE):** Each PE firm observes the full realized intermediate state $\tilde{s}$ and solves a binary logit $\{\text{enter}, \text{quit}\}$. We compute CCPs for all intermediate states reachable from s_0.
   - **Stage 3 (new):** Each new firm solves a binary logit $\{\text{stay}, \text{exit}\}$, conditioning on pre-computed PE CCPs to integrate over the PE-stage continuation value.
   - **Stage 2 (both):** Each both firm solves a binary logit, conditioning on new + PE CCPs. Innovators locked in from Stage 1 (k_{io} old firms that chose innovate) are added to n_b before this stage.
   - **Stage 1 (old):** Each old firm solves a multinomial logit $\{\text{stay}, \text{innovate}, \text{exit}\}$, conditioning on all lower-stage CCPs.

   No fixed-point iteration is needed within the period: the sequential structure makes each stage a closed-form single-agent problem given pre-computed downstream CCPs. Marginal CCPs for later-moving types are recovered by integrating over the equilibrium transition distribution (products of independent Binomial / Multinomial draws).

3 Calibration

We summarize the parameters we use for calibration and for simulating the data in Table 1. We set $\kappa = \phi = 0.2$ so that any remaining structural difference between incumbents and entrants in their period-1 adoption decision is driven by **cannibalization** alone: a "both"-type firm internalizes that adding new-gen output erodes its own old-gen profit, while a

PE entrant has no legacy old-gen output to displace. The two policy counterfactuals below (Sections 6 and 7) lean on this distinction: the innovation subsidy pulls the cannibalization-discounted margin, while the entry subsidy pulls the clean margin.

Table 1: Parameter values for calibration and simulation

Parameter	Description	Value
A	Demand intercept	3.0
B	Demand slope	1.0
M	Market size	1.0
c_o	Old-tech marginal cost	1.5
$c_{n,0}$	New-tech baseline cost	0.5
β	Discount factor	0.9
κ	Innovation cost	0.2
ϕ	Entry cost	0.2
σ	EVT1 scale	0.5
γ	Agglomeration (default)	0.15
ρ	Cross-market substitution	0.5

4 Regional Extension

4.1 Regional Structure

To study the interaction between industrial policy and agglomeration we extend the state space with a **region** attribute. There are $R = 3$ regions (labelled $r \in \{1, 2, 3\}$); each firm is born into a region and never leaves it, and potential entrants sit in per-region pools $n_{pe,r}$. Product-market competition stays **global**, with all active firms meeting in the same two-market Cournot game of equations (1)–(2). Agglomeration spillovers, in contrast, are now **local**:

$$c_{n,r,t} = c_{n,0} - \gamma_r \cdot \big(n_{b,r,t} + n_{n,r,t}\big), \qquad r = 1, \ldots, R. \tag{6}$$

Because $c_{n,r,t}$ varies across regions, firms of type "both" and "new" face *heterogeneous* marginal costs in the new-gen market. The Cournot stage therefore loses symmetry across regions, and equilibrium quantities are obtained by solving an asymmetric FOC system of up to $4R$ equations per state. Within-region slots remain symmetric (all firms in the same $(type, r)$ cell pick the same quantity), so per-firm profits are still characterised by $\pi_{o,r} = bq^2_{oo,r}$, $\pi_{n,r} = bq^2_{nn,r}$, and $\pi_{b,r} = b(q^2_{bo,r} + q^2_{bn,r}) + 2s\, q_{bo,r} q_{bn,r}$ where $b = B/M$ and $s = \rho B/M$.

Each firm privately captures the lower cost when it innovates but does not internalize the benefit it confers on regional peers. This is a standard **local agglomeration externality**, creating a wedge between the social and private return to innovation and opening the door to regionally targeted industrial policy.

4.2 Solution with Regional States

The sequential-move structure (old $\to$ both $\to$ new $\to$ pe) carries over unchanged, and we extend the same idea one level down: **within each stage, regions move sequentially**, $r = 1 \to 2 \to 3$. When firms in region r decide, they observe the realized outcome of regions $1, \ldots, r-1$ and integrate over the (yet-to-play) strategies of regions $r+1, \ldots, R$. This is the regional analogue of Igami's cross-type sequencing: it makes every sub-stage a single-agent problem whose CCP is a *scalar* fixed point rather than a joint fixed point across regions, and it removes any coupling in the solver between regions at the same stage.

Concretely, we solve each stage in reverse region order. Region R's scalar FP is computed first for every reachable sub-state, and cached. Region $R-1$'s FP is then computed given the cached region-R continuation, and so on down to region 1. Across stages, the usual layered caches still apply: stage-4 (PE) depends only on the post-new-stage state and is reused across initial states via $\mathrm{EV}_{\text{after pe}}[s_{an}, r]$; stages 3, 2, 1 cascade through $\mathrm{EV}_{\text{after new}}$, $\mathrm{EV}_{\text{after both}}$, and $\mathrm{EV}_{\text{after old}}$, each keyed by (State, r). Because each sub-state's CCP is a function of the realized history, the per-region CCPs reported in the results below are *marginalised* over the equilibrium play of earlier-moving regions, computed by a forward traversal of the sub-stage tree.

Because the state space now has $4R = 12$ integer coordinates, we cap $N_{\max} = 6$ on this branch (about 1.8×10^4 enumerated states) and pick initial states with total firm count ≤ 6. One visible artefact of the sequential-move convention is a small first-mover asymmetry: at any symmetric s_0 with uniform γ_r, the three regions' CCPs differ by at most 10^{-3}. Region 1 commits without observing anyone, while region 3 best-responds to the realised history. The gap is well below the magnitude of every counterfactual difference reported in Section A.

5 Estimation on Simulated Data

As a first pass at recovering the model's structural parameters from data, we simulate a firm-level dataset from the regional model under the calibration of Table 1 and then estimate (γ, κ, ϕ), treating the demand parameters, the EVT1 scale σ, the cost $c_{n,0}$, and the state cap $N_{\max}$ as known.

5.1 Simulated dataset

The simulator (`code/scripts/simulate_data.jl`) draws 500 independent markets from the 2-period regional model. Each market gets its own initial state $s_0 \sim \mathrm{unif}\{0, 1, 2\}^{4R}$ (rejecting empty or over-capacity draws), so the panel covers a range of local cluster sizes. For every firm in every period we record region, firm type, the two Cournot quantities, and realised profit. For period 1 we additionally record the discrete action. The structural parameters used to draw the data are deliberately *not* stored in the CSV.

5.2 Two-step estimator

Because equilibrium quantities are a deterministic function of the state and the spillover vector γ, identification splits cleanly: γ is pinned by the Cournot block and (κ, ϕ) by the discrete actions, conditional on γ. We exploit this with a two-step estimator (see `notes/estimation.md` for the full writeup).

Step 1: NLS for γ. For each market we reconstruct s_0 and s_1 by counting firms of each type and region in the corresponding period. We then solve

$$\widehat{\gamma} \;=\; \arg\min_{\gamma} \sum_{m,t,r,\,\mathrm{slot}} \big(q^{\mathrm{obs}}_{m,t,r,\mathrm{slot}} - \hat{q}_{m,t,r,\mathrm{slot}}(\gamma)\big)^2, \tag{7}$$

starting from $\gamma^0 = (0, 0, 0)$ and leaving γ *unconstrained*. Because the simulated quantities are noiseless, the minimum is machine zero (SSR $= 4.24e-13$) and $\widehat{\gamma}$ equals the truth to all reported digits; this step doubles as a sanity check that state reconstruction is correct.

Step 2: MLE for (κ, ϕ). Holding $\gamma = \widehat{\gamma}$ fixed, we maximise the period-1 action log-likelihood over (κ, ϕ). The likelihood *replays* the sequential-move order used by the solver: within each stage, regions act in order $r = 1 \to 2 \to 3$, and each firm's contribution uses the stage CCP evaluated at the sub-state its firm actually faced (not the marginal CCP integrated over the forward traversal). Per market,

$$\begin{aligned}
\ell_m(\kappa, \phi) = &\sum_{r=1}^{R} \Big[k_r^{\mathrm{so}} \log p_{\mathrm{so}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{io}} \log p_{\mathrm{io}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{eo}} \log\big(1 - p_{\mathrm{so}} - p_{\mathrm{io}}\big) \Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{sb}} \log p_{\mathrm{sb}}(\tilde{s}_r^{\mathrm{BOTH}}) + k_r^{\mathrm{eb}} \log\big(1 - p_{\mathrm{sb}}\big) \Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{sn}} \log p_{\mathrm{sn}}(\tilde{s}_r^{\mathrm{NEW}}) + k_r^{\mathrm{en}} \log\big(1 - p_{\mathrm{sn}}\big) \Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{ep}} \log p_{\mathrm{ep}}(\tilde{s}_r^{\mathrm{PE}}) + k_r^{\mathrm{op}} \log\big(1 - p_{\mathrm{ep}}\big) \Big],
\end{aligned} \tag{8}$$

where the $k_r^{\bullet}$ are observed action counts in region r and the $\tilde{s}_r^{\bullet}$ are the sub-states produced by sequentially updating s_0 with the realised outcomes of earlier regions and stages. The total objective is $\ell(\kappa, \phi) = \sum_m \ell_m(\kappa, \phi)$, maximised by Nelder-Mead starting from $(\kappa^0, \phi^0) = (1.0, 1.0)$. The starting point is intentionally far from the calibration, so the optimiser must traverse the parameter space. Standard errors come from the BHHH estimator $V = (\sum_m g_m g_m')^{-1}$ with central finite-difference gradients.

5.3 Results

Table 2 reports the estimates against the data-generating parameters. The two-step estimator recovers $\widehat{\gamma} = (0.1500, 0.1500, 0.1500)$ exactly (Step 1), and the action MLE delivers

$\widehat{\kappa} = 0.2193$ (s.e. 0.0402, truth 0.2000) and $\widehat{\phi} = 0.1810$ (s.e. 0.0403, truth 0.2000). Both κ and ϕ lie within roughly one standard error of their true values; given 500 markets and a handful of deciders per market, this is about the precision the sample supports. The period-1 action log-likelihood at the optimum is -1852.123.

Table 2: Two-step estimation of (γ, κ, ϕ)

Parameter	Truth	Estimate	S.E.
κ	0.2000	0.2193	0.0402
ϕ	0.2000	0.1810	0.0403
γ_1	0.1500	0.1500	—
γ_2	0.1500	0.1500	—
γ_3	0.1500	0.1500	—

Notes: Estimates from 500 simulated markets. Step 1 identifies γ from Cournot quantities; Step 2 estimates (κ, ϕ) by period-1 action MLE conditional on $\widehat{\gamma}$. Standard errors (BHHH) are reported only for the MLE step.

5.4 Strawman: a misspecified model without agglomeration

What if an analyst sees the same Cournot data but never considers agglomeration? They attribute cross-region cost differences to exogenous regional fundamentals, factor prices and geography, and estimate a region intercept on new-tech marginal cost while imposing $\gamma \equiv 0$. The misspecified cost function is

$$c_{n,r}^{\text{mis}} = c_{n0} + \delta_r,$$

state-independent within a region. Step 1 NLS on the same Cournot data pins $\widehat{\delta}_r$ near $-\widehat{\gamma}_r \cdot \mathbb{E}[n_b^r + n_n^r]$ and leaves a residual SSR of 18.27, against essentially zero for the correctly-specified Step 1. Step 2 MLE on period-1 actions then yields $\widetilde{\kappa} = 0.2078$ and $\widetilde{\phi} = 0.1697$, both within a few percentage points of $\widehat{\kappa}$ and $\widehat{\phi}$.

The misspecified model fits action margins by construction but mispredicts heterogeneity and comparative statics. Table 3 compares the two models on three blocks of moments evaluated at the period-0 states observed in the data. The innovation-rate panel is essentially a tie: both models match the average $P(\text{innov} \mid \text{old})$ and $P(\text{enter} \mid \text{pe})$ to four decimals, so the innovator's-dilemma gap is identical. This is a feature of MLE. Action frequencies are pinned at the realized states. The Lerner panel and the comparative-static panel are where the strawman breaks down.

Markup heterogeneity. On the new-generation side the misspecified model under-predicts both the level and the dispersion of markups: mean Lerner index 0.7212 versus 0.7574, standard deviation 0.0575 versus 0.0713. By assumption the misspecified marginal cost does not move with peer density, so realized markups do not either. The true model captures the

cross-state covariance between cost and quantity that the misspecified one collapses.

Comparative-static prediction. A counterfactual +1 `both`-firm shock to region 1 produces $\Delta P(\text{innov} \mid \text{old}, r = 1) = -0.0402$ in the misspecified model and -0.0139 in the true model (firm-weighted across the period-0 states where the perturbation is feasible). The drop is roughly three times larger under misspecification. A misspecified analyst would conclude that peer entry sharply discourages further innovation. The true story is much milder, because the cost-reduction channel partially offsets the negative price effect of an additional new-generation supplier. The gap between the two values is a one-number summary of the agglomeration externality the strawman is missing.

Table 3: Correct vs. misspecified model: estimates and predictions

	True ($\widehat{\gamma} > 0$)	Misspec. ($\gamma \equiv 0$)
Estimated parameters		
$\widehat{\kappa}$	0.2193	0.2078
$\widehat{\phi}$	0.1810	0.1697
$\widehat{\gamma}_1$	0.1500	—
$\widehat{\gamma}_2$	0.1500	—
$\widehat{\gamma}_3$	0.1500	—
$\widehat{\delta}_1$	—	-0.2312
$\widehat{\delta}_2$	—	-0.2319
$\widehat{\delta}_3$	—	-0.2250
Period-1 actions (firm-weighted across period-0 states)		
$P(\text{innov} \mid \text{old})$	0.3795	0.3798
$P(\text{enter} \mid \text{pe})$	0.5624	0.5622
Innovator's-dilemma gap	0.1828	0.1825
Lerner index $(P - MC)/P$ (across-market mean / s.d.)		
Old generation	0.1470 / 0.0497	0.1494 / 0.0478
New generation	0.7574 / 0.0713	0.7212 / 0.0575
Comparative static: response to *a +1 BOTH-firm shock to region 1 (firm-weighted)*		
$\Delta P(\text{innov} \mid \text{old}, r = 1)$	-0.0139	-0.0402

Notes: The misspecified model imposes $\gamma \equiv 0$ and fits a region intercept δ_r on new-tech marginal cost. Step 1 re-estimates $(\delta_1, \delta_2, \delta_3)$ by NLS on observed Cournot quantities; Step 2 re-estimates (κ, ϕ) by MLE on period-1 actions, conditional on the misspecified Cournot continuation values. Innovation-rate moments are firm-weighted across (market, region) at the period-0 states observed in `data/simulated_data.csv`. Lerner indices $(P - MC)/P$ are quantity-weighted within a market and equal-weighted across markets, with the across-market standard deviation reported after the slash. The comparative static $\Delta P(\text{innov} \mid \text{old}, r = 1)$ averages, firm-weighted across markets, the response of the region-1 innovation rate to adding one `both` firm in region 1; markets where the perturbation would exceed $N_{\max}$ are dropped.

6 Counterfactual: Region-3 Innovation Subsidy

A region-3 innovation subsidy of size τ shifts the *private* cost of innovation in region 3 from $\widehat{\kappa}$ to $(\widehat{\kappa} - \tau)$ while leaving the social resource cost unchanged at $\widehat{\kappa}$. Sweeping τ over the grid $\tau/\widehat{\kappa} \in \{0, 0.10, \ldots, 0.50\}$ produces a clean planner-vs-country split: region 3's own welfare is weakly *increasing* in τ, but aggregate welfare across the three regions has a small interior optimum at $\tau/\widehat{\kappa} = 0.30$ ($\Delta\Sigma W = +0.00\,\%$). At the largest grid point $\tau/\widehat{\kappa} = 0.50$, region 3's $P(\text{innov} \mid \text{old})$ rises by $+14.06\,\%$ and its welfare gains $+0.12\,\%$, but regions 1 and 2 lose $|-0.05|\,\%$ and $|-0.07|\,\%$, so the planner stops well short of the corner. The wedge between region 3's preferred subsidy and the planner's is the cross-region Cournot externality: the subsidy makes region 3's new-tech producers cheaper in a globally competitive product market, and the producers in regions 1 and 2 lose share. The rest of the section sets up the subsidy specification, derives the welfare formula under sovereign funding, describes the Monte Carlo evaluation, and walks through the mechanism.

6.1 Subsidy specification

A region-r innovation subsidy of size $\tau_r \geq 0$ changes only the *private* payoff of an old-tech firm switching to "both" technology:

$$u_r^{\text{innov}} = \pi_r^{\text{both}}(s) - (\kappa - \tau_r) + \beta\, \text{EV}_r^{\text{old}\to\text{both}}, \tag{9}$$

leaving the entry decision and the rest of the game unchanged. The implementation adds a single `subsidy` field to the parameter struct; the default `(0,0,0)` recovers the no-subsidy baseline. We sweep the one-region policy $\tau = (0, 0, \tau)$ to isolate the question "what does targeted industrial policy in region 3 do?" from the broader question of a global R&D subsidy. The spillover specification stays at the local-only baseline of equation (6).

6.2 Welfare under sovereign funding

Each region is treated as a sovereign country: its own government funds its own subsidy from its own taxpayers, so the transfer $\tau_r \cdot k_r^{\text{innov}}$ flows from region r's taxpayers to region r's firms and cancels exactly inside region r's welfare. The *social* resource cost remains the full κ even when the firm only pays $(\kappa - \tau_r)$; the subsidy reduces the private burden, not the real labour and materials embodied in a new-technology blueprint. Two-period discounted welfare per region is therefore

$$W_r = \tfrac{1}{R}\big[\text{CS}(s_0) + \beta\,\text{CS}(s_1)\big] + \text{PS}_r(s_0) + \beta\,\text{PS}_r(s_1) - \kappa\, k_r^{\text{innov}} - \phi\, k_r^{\text{enter}}, \tag{10}$$

identical in form to equation (15): the subsidy is a within-region transfer that does not appear in W_r. With this accounting, the identity $\sum_r W_r = \text{CS}_{\text{total}} + \sum_r \text{PS}_r - \sum_r (\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}})$ holds exactly in the baseline and in every grid scenario. The runner verifies it numerically and finds residuals at machine precision.

6.3 Evaluation by Monte Carlo with common random numbers

We average $K = 5000$ independent simulations per scenario. Each market draws its own initial state s_0 from the same distribution `simulate_data.jl` uses, so the counterfactual sits on the same population the model was estimated on. The same seed (20260424) drives the baseline and every grid point, with per-market `MersenneTwister(seed + k)`. Identical s_0 draws and identical EVT1 shock paths under all scenarios mean each market contributes paired observations $(W_r^{\text{base}}, W_r^{\tau})$, so the variance of the welfare difference is much smaller than the variance of either level. Headline numbers stabilise to roughly 0.05 percentage points by $K = 1000$.

6.4 Calibration and results

We hold the demand parameters, β, σ, ρ, c_o, $c_{n,0}$, and $N_{\max}$ at the values of Table 1, and use the structural estimates from Section 5: $\widehat{\kappa} = 0.2193$, $\widehat{\phi} = 0.1810$, and uniform $\widehat{\gamma}_r = 0.1500$.

Table 4 reports the full grid sweep. Region 3's innovation rate climbs steadily from 0.3860 at $\tau = 0$ to 0.4402 at the largest grid point, while regions 1 and 2 dip slightly through equilibrium feedback. Aggregate welfare is non-monotonic in τ: it rises modestly with the first units of subsidy, peaks at $\tau/\widehat{\kappa} = 0.30$, and declines beyond that as cross-region producer-surplus losses outweigh the additional own-region gain.

Table 4: Region-3 innovation subsidy: grid sweep over $\tau/\widehat{\kappa}$

$\tau/\widehat{\kappa}$	τ	$P_{\text{innov},1}$	$P_{\text{innov},2}$	$P_{\text{innov},3}$	ΔW_1	ΔW_2	ΔW_3	$\Delta\Sigma W$	$\Delta\Sigma W$ (%)
0.00	0.0000	0.3846	0.3779	0.3860	+0.0000	+0.0000	+0.0000	+0.0000	+0.00%
0.10	0.0219	0.3842	0.3779	0.3921	-0.0001	-0.0000	+0.0003	+0.0001	+0.00%
0.20	0.0439	0.3837	0.3775	0.3996	-0.0002	-0.0004	+0.0007	+0.0001	+0.00%
0.30	0.0658	0.3837	0.3775	0.4164	-0.0007	-0.0008	+0.0017	+0.0002	+0.00%
0.40	0.0877	0.3837	0.3775	0.4292	-0.0009	-0.0013	+0.0023	+0.0000	+0.00%
0.50	0.1096	0.3837	0.3770	0.4402	-0.0011	-0.0014	+0.0025	+0.0000	+0.00%

Notes: Monte Carlo over $K = 5000$ independent markets, each drawing its own s_0 from the DGP used by `simulate_data.jl`; common random numbers across all scenarios (seed = 20260424). Calibration: $\widehat{\kappa} = 0.2193$, $\widehat{\phi} = 0.1810$, uniform $\widehat{\gamma}_r = 0.1500$. The subsidy is region-3 only: $(0, 0, \tau)$. Welfare is computed under sovereign funding, so the transfer cancels within region 3. CS allocated equally across the three regions; PS_r aggregates own-region firms' profits in both periods; costs paid charges the full $\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}}$ to region r.

6.5 Discussion

Three points are worth highlighting.

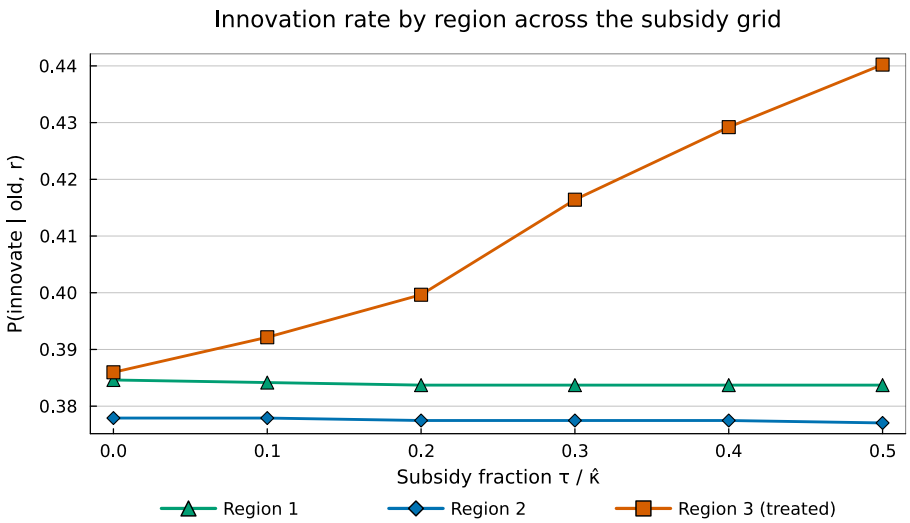

Figure 1: Innovation rate $P(\text{innov} \mid \text{old}, r)$ by region across the subsidy grid. Only region 3 (treated) responds strongly; regions 1 and 2 dip slightly through equilibrium feedback.

The subsidy works on its target margin. Region 3's $P(\text{innov} \mid \text{old})$ rises from 0.3860 to 0.4402 at $\tau/\widehat{\kappa} = 0.50$, a $+14.06\,\%$ lift. The mechanism is direct: a lower private cost moves the threshold draw of the EVT1 shock that triggers innovation. As a policy lever for raising regional R&D activity, the subsidy works.

Region 3 alone wants the subsidy maxed; the planner does not. A subsidy is welfare-improving for the planner only if there is a positive externality on the subsidised margin that the firm fails to internalise. In our calibration the agglomeration externality $\widehat{\gamma}_r = 0.1500$ is *local*: region 3's innovation today helps only region 3's own period-2 cost, and the firm already internalises the bulk of that benefit through its expected period-2 profit. Cournot competition across regions, in contrast, makes region-3 expansion a *negative* externality on regions 1 and 2, which the planner does internalise. Region 3 alone sees only the gain and prefers $\tau/\widehat{\kappa} = 0.50$; the planner balances the gain against the cross-region loss and settles on the interior $\tau/\widehat{\kappa} = 0.30$.

The non-treated regions bear small but consistent losses. Cheaper region-3 production drags the global new-technology price down, so regions 1 and 2 see lower firm profits. The CS share they receive from cheaper goods is too small to compensate. At $\tau/\widehat{\kappa} = 0.50$, region 1 loses $|-0.05|\,\%$ of its baseline welfare and region 2 loses $|-0.07|\,\%$. These are small but signed in the same direction across all $K \in \{500, 1000, 5000\}$.

The result is calibration-dependent. Under a spillover specification that pools region 3 with another region (an alliance variant of Appendix A), the planner would internalise the spillover from region-3 innovation onto its partner and the optimum would shift right. With a larger $\widehat{\gamma}$ or a tighter ρ, the cross-region effects scale accordingly. Full details, including the K-stability

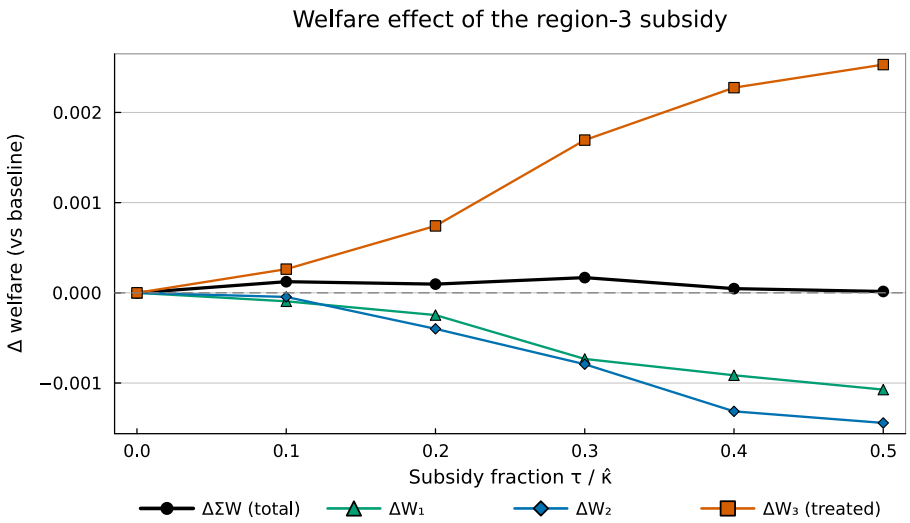

Figure 2: Per-region and total welfare change vs. baseline as a function of $\tau/\widehat{\kappa}$. Region 3's own welfare is weakly increasing across the grid, while regions 1 and 2 lose. Aggregate welfare (black) peaks at the interior point $\tau/\widehat{\kappa} = 0.30$.

diagnostic and the sovereign-funding accounting, are in `notes/subsidy_counterfactual.md`.

7 Counterfactual: Region-3 Entry Subsidy

A region-3 *entry* subsidy of size ψ shifts the private cost of entering as a new-technology firm in region 3 from $\widehat{\phi}$ to $(\widehat{\phi} - \psi)$ while leaving the social resource cost of entry unchanged at $\widehat{\phi}$. The same planner-vs-country split appears as in the innovation case, but at a different optimum: region 3 alone prefers $\psi/\widehat{\phi} = 0.50$ (its welfare rises by $+0.15\,\%$ at the corner), while the planner picks the interior $\psi/\widehat{\phi} = 0.20$ ($\Delta\Sigma W = +0.00\,\%$). At the corner, regions 1 and 2 lose $|-0.09|\,\%$ and $|-0.06|\,\%$ respectively, so the cross-region loss eats up the own-region gain.

The entry-subsidy story differs from the innovation-subsidy story in two structural ways, both visible in the numbers. First, region 3's own welfare gain at $\psi/\widehat{\phi} = 0.5$ is $+0.15\,\%$ versus $+0.12\,\%$ under the innovation subsidy — because new entrants add new-tech quantity without cannibalising existing old-gen revenue, the own-region pass-through is larger per dollar of subsidy. Second, the larger pass-through also feeds back into larger cross-region losses, so the planner's optimal entry subsidy (0.20) is *smaller* than its optimal innovation subsidy (0.30): the cleaner instrument also has a steeper externality cost, and the planner stops earlier on the entry axis.

7.1 Subsidy specification

A region-r entry subsidy of size $\psi_r \geq 0$ changes only the *private* payoff of a potential entrant becoming a new-tech firm:

$$u_r^{\text{enter}} = -(\phi - \psi_r) + \beta\, \text{EV}_r^{\text{pe}\to\text{new}}, \tag{11}$$

leaving the innovation decision and the rest of the game unchanged. The implementation adds a single `entry_subsidy` field to the parameter struct; the default `(0,0,0)` recovers the no-subsidy baseline. We sweep the one-region policy $\psi = (0, 0, \psi)$ to isolate targeted entry policy in region 3 from the question of a global entry subsidy. The spillover specification stays at the local-only baseline.

7.2 Welfare under sovereign funding

As in the innovation-subsidy section, region r's government funds its own subsidy from its own taxpayers, so the transfer $\psi_r \cdot k_r^{\text{enter}}$ flows from region r's taxpayers to region r's entrants and cancels exactly inside region r's welfare. The social resource cost of an entry remains the full ϕ. Two-period discounted welfare per region is

$$W_r = \tfrac{1}{R}\big[\text{CS}(s_0) + \beta\, \text{CS}(s_1)\big] + \text{PS}_r(s_0) + \beta\, \text{PS}_r(s_1) - \kappa\, k_r^{\text{innov}} - \phi\, k_r^{\text{enter}}, \tag{12}$$

identical in form to equation (15): the entry subsidy is a within-region transfer that does not appear in W_r. The identity $\sum_r W_r = \text{CS}_{\text{total}} + \sum_r \text{PS}_r - \sum_r (\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}})$ holds at machine precision in baseline and in every grid scenario.

7.3 Evaluation by Monte Carlo with common random numbers

We average $K = 5000$ independent simulations per scenario, with the same seed (20260424) across baseline and every grid point and per-market `MersenneTwister(seed + k)`. Headline numbers stabilise to roughly 0.05 percentage points by $K = 1000$; the aggregate welfare effect is small (about a basis point) but its sign is robust across K.

7.4 Calibration and results

We use the structural estimates from Section 5: $\widehat{\kappa} = 0.2193$, $\widehat{\phi} = 0.1810$, and uniform $\widehat{\gamma}_r = 0.1500$.

Table 5 reports the full grid sweep. Region 3's entry rate climbs from 0.5757 at $\psi = 0$ to 0.6163 at the largest grid point, while regions 1 and 2 dip slightly through equilibrium feedback. Region 3's own welfare is weakly increasing across the grid, but aggregate welfare peaks at the interior point $\psi/\widehat{\phi} = 0.20$ and turns negative at higher ψ as cross-region losses grow.

Table 5: Region-3 entry subsidy: grid sweep over $\psi/\widehat{\phi}$

$\psi/\widehat{\phi}$	ψ	$P_{\text{enter},1}$	$P_{\text{enter},2}$	$P_{\text{enter},3}$	ΔW_1	ΔW_2	ΔW_3	$\Delta\Sigma W$	$\Delta\Sigma W$ (%)
0.00	0.0000	0.5544	0.5766	0.5757	+0.0000	+0.0000	+0.0000	+0.0000	+0.00%
0.10	0.0181	0.5539	0.5766	0.5852	-0.0003	-0.0003	+0.0007	+0.0001	+0.00%
0.20	0.0362	0.5539	0.5766	0.5943	-0.0009	-0.0007	+0.0018	+0.0002	+0.00%
0.30	0.0543	0.5535	0.5766	0.6051	-0.0015	-0.0010	+0.0026	+0.0001	+0.00%
0.40	0.0724	0.5535	0.5766	0.6090	-0.0016	-0.0011	+0.0029	+0.0002	+0.00%
0.50	0.0905	0.5530	0.5766	0.6163	-0.0019	-0.0013	+0.0033	+0.0001	+0.00%

Notes: Monte Carlo over $K = 5000$ independent markets, each drawing its own s_0 from the DGP used by `simulate_data.jl`; common random numbers across all scenarios (seed = 20260424). Calibration: $\widehat{\kappa} = 0.2193$, $\widehat{\phi} = 0.1810$, uniform $\widehat{\gamma}_r = 0.1500$. The subsidy is region-3 only: $(0, 0, \psi)$. Welfare is computed under sovereign funding, so the transfer cancels within region 3.

7.5 Discussion

Three points are worth highlighting.

Entry pulls a cleaner margin than innovation. Innovating firms are old-tech incumbents that switch to "both". They were already producing old-tech output, and the new-tech output they add as "both" partly cannibalises that old-tech revenue through cross-market substitution ρ. New entrants, in contrast, arrive with no legacy old-tech output to displace. Each marginal entry therefore adds more total new-tech quantity at the global Cournot stage than each marginal innovation does. With $\kappa = \phi$ in our calibration, the per-dollar private-cost reduction is the same under the two policies, so the difference shows up directly in region 3's welfare gain: at the corner, +0.15 % under entry versus +0.12 % under innovation. The cross-region losses scale up correspondingly: $|-0.09|$ % for region 1 versus $|-0.05|$ % under innovation.

The planner wants both subsidies, but at modest interior levels. Under the innovation subsidy alone the planner's optimum is $\tau/\widehat{\kappa} = 0.30$ ($\Delta\Sigma W = +0.00$ %); under the entry subsidy alone it is $\psi/\widehat{\phi} = 0.20$ ($\Delta\Sigma W = +0.00$ %). The entry-subsidy optimum is smaller in absolute terms because the entry margin has higher pass-through and therefore a steeper marginal externality cost on regions 1 and 2: the planner accepts the cleaner instrument at a smaller dose. In both cases, region 3 alone would prefer the corner — the wedge between its preferred subsidy and the planner's is exactly the cross-region externality that the planner internalises and region 3 does not.

The aggregate effects are small but signed. Both interior optima deliver a $\Delta\Sigma W$ on the order of 0.01 % of baseline. This is small in level, but the interior optimum is what makes the planner-vs-region-3 split well-defined: region 3 has a strictly preferred policy (the corner) and the planner has a strictly different one (the interior). The joint sweep below sharpens this contrast on the two-instrument grid.

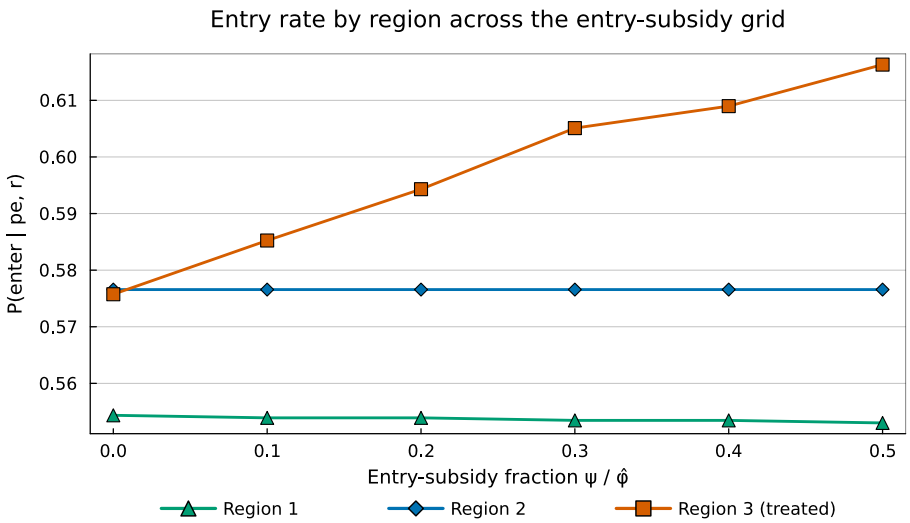

Figure 3: Entry rate $P(\text{enter} \mid \text{pe}, r)$ by region across the entry-subsidy grid. Only region 3 (treated) responds strongly; regions 1 and 2 dip slightly through equilibrium feedback.

7.6 Joint subsidy grid

A joint sweep over (τ, ψ) on the 6×6 grid lets the planner trade the two instruments off against each other. We use it to make three points, which the heatmap of Figure 5 summarises. The two markers on the heatmap correspond to the planner's argmax under the two calibrations: the gold star is the agglomeration calibration $\gamma = \widehat{\gamma}$; the white diamond is the no-agglomeration counterfactual $\gamma = 0$.

(1) The planner's optimal subsidy is asymmetric: more entry than innovation. At the agglomeration calibration the planner picks $(\tau/\widehat{\kappa}, \psi/\widehat{\phi}) = (0.10, 0.20)$, with the entry-subsidy fraction substantially larger than the innovation-subsidy fraction. The mechanism is the cannibalization wedge of Igami's "innovator's dilemma": an old-tech incumbent who switches to "both" adds new-tech output but partly cannibalises its own old-tech revenue, so it produces *less* new-tech quantity than a fresh entrant of equivalent type. A unit of entry subsidy therefore moves the new-tech market further than a unit of innovation subsidy. The planner internalises this and tilts the mix toward entry; the own-region welfare-per-corner numbers in Sections 6 and 7 confirm this (+0.15 % for entry vs. +0.12 % for innovation at $\psi/\widehat{\phi} = \tau/\widehat{\kappa} = 0.5$).

(2) Agglomeration is the externality the planner is correcting. Under $\gamma = \widehat{\gamma} = 0.1500$, the local agglomeration spillover is a positive externality: adding an innovator (or entrant, since both end up using new tech) lowers next period's $c_{n,r}$ for every other new-tech firm in region r. The firm internalises only its own period-2 cost reduction; the rest of the

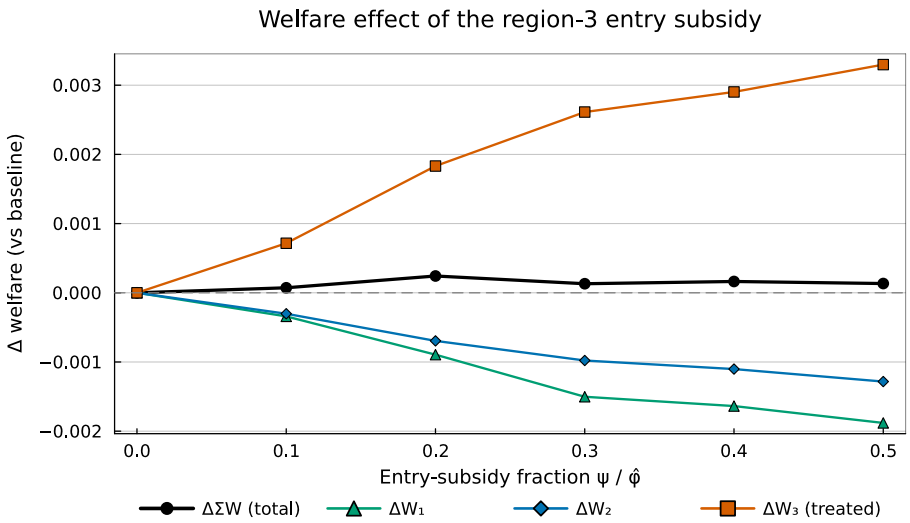

Figure 4: Per-region and total welfare change vs. baseline as a function of $\psi/\widehat{\phi}$. Region 3's own welfare rises monotonically with ψ; regions 1 and 2 lose. Aggregate welfare (black) peaks at the interior point $\psi/\widehat{\phi} = 0.20$ and turns negative at higher ψ.

cluster gets it for free. A subsidy that nudges $P(\text{innov})$ or $P(\text{enter})$ above the laissez-faire level helps close the gap. With γ shut off, that channel disappears and only the *negative* cross-region Cournot externality remains. The planner then prefers *no* subsidy at all: the no-agglomeration argmax is the corner $(\tau/\widehat{\kappa}, \psi/\widehat{\phi}) = (0.00, 0.00)$, with $\Delta\Sigma W = +0.00\,\%$, and every positive cell in the no-agglomeration grid is welfare-decreasing (largest cell $-0.09\,\%$). The size of the optimal subsidy is therefore proportional to the strength of the agglomeration externality, not a robust feature of the model.

(3) A unilateral region-3 government over-subsidises. Region 3's own welfare is monotone non-decreasing across the entire (τ, ψ) grid: it picks the corner $(0.50, 0.50)$, where $\Delta W_3 = +0.27\,\%$. The planner stops far short of the corner because the planner internalises the producer-surplus losses imposed on regions 1 and 2 by cheaper region-3 output and region 3 does not. The wedge between region 3's preferred policy and the planner's is exactly the cross-region Cournot terms-of-trade externality. It would shrink if the spillover specification pooled region 3 with another region, since the planner would then internalise more of the spillover gain to region 3's partner (cf. the alliance variant of Appendix A). Full details, including the K-stability diagnostic, are in `notes/entry_subsidy_counterfactual.md`.

8 Next Steps

1. **Heterogeneous absorptive capacity.** The alliance result here uses uniform $\widehat{\gamma}_r$. Allowing γ_r to differ across regions, with the bloc-pooled spec of equation (13), would let one region's higher absorptive capacity amplify the alliance's effect.

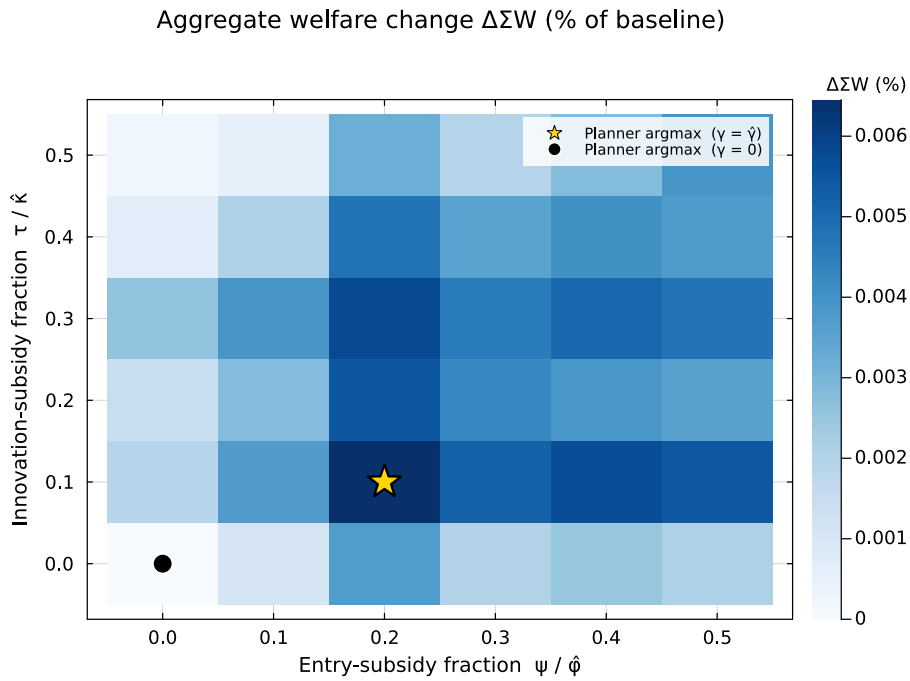

Figure 5: Aggregate welfare change $\Delta\Sigma W$ (% of baseline) on the joint $(\tau/\widehat{\kappa}, \psi/\widehat{\phi})$ grid, evaluated at $\gamma = \widehat{\gamma}$. The *gold star* marks the planner's argmax under $\gamma = \widehat{\gamma}$: $(0.10, 0.20)$, with $\Delta\Sigma W = +0.01\,\%$. The *white diamond* marks the planner's argmax in the same problem with γ set to zero: $(0.00, 0.00)$. Region 3's own-welfare argmax is the south-east corner of the grid $(0.50, 0.50)$, where $\Delta W_3 = +0.27\,\%$.

2. **Fiscal cost of policy.** The current welfare accounting prices in firms' own costs paid but does not net out the fiscal cost of any subsidy a government would need to deliver to enlarge γ_r in a target region. Adding government revenue and cost makes the targeted-subsidy and alliance scenarios directly comparable as policy levers.

3. $T > 2$ **horizon.** A genuine multi-period extension would force the value-function recursion to use the $\sigma \log \sum \exp$ ex-ante form at every internal period. The current 2-period setup avoids this because period 2 is terminal.

A Counterfactual: EU–US Alliance

Pooling regions 1 and 2's innovator counts lifts welfare by roughly $+6.46\,\%$ in each allied region, leaves region 3 unchanged in welfare terms, and raises aggregate welfare by $+4.31\,\%$ of baseline. Innovation rates rise by $+4.71\,\%$ and $+6.14\,\%$ in the allied regions and fall by $|-3.89|\,\%$ in region 3, so the unallied region is harmed on the real-economy margin even though it breaks even on welfare. The rest of the section sets up the spillover specification, derives the welfare formula, describes the Monte Carlo evaluation, and walks through the mechanism.

A.1 Spillover specification under the alliance

Under the baseline, region r's new-technology marginal cost depends only on its own innovators, equation (6). Under the alliance of $\{1, 2\}$, regions 1 and 2 share an enlarged pool:

$$c_{n,r,t}^{\text{alliance}} = \max\left\{0,\ c_{n,0} - \gamma_r \cdot \sum_{r' \in \text{bloc}(r)} \left(n_{b,r',t} + n_{n,r',t}\right)\right\}, \qquad \text{bloc}(1) = \text{bloc}(2) = \{1,2\},\ \text{bloc}(3) = \{3\}. \tag{13}$$

The cost floor prevents the marginal cost from turning negative; under the calibration $\gamma \leq 0.15$ this constraint never binds. The implementation adds a single `blocs` field to the parameter struct; the default `(1,2,3)` recovers equation (6) bit-for-bit.

A.2 Welfare

The two-market linear demand of equations (1)–(2) is rationalised by the quasi-linear utility $U = A(Q_o + Q_n) - \frac{b}{2}(Q_o^2 + Q_n^2) - s\,Q_o\,Q_n$ with $b = B/M$ and $s = \rho B/M$, giving consumer surplus

$$\text{CS}(s) = \tfrac{b}{2}(Q_o^2 + Q_n^2) + s\,Q_o\,Q_n. \tag{14}$$

Because Cournot is global, CS is a single number; we allocate it equally across regions (equal-population assumption). Producer surplus is local: each firm's per-period profit is charged to its own region. Innovation and entry costs are paid by the firm's region in period 1. Two-period discounted welfare per region is

$$W_r = \tfrac{1}{R}\big[\text{CS}(s_0) + \beta\,\text{CS}(s_1)\big] + \text{PS}_r(s_0) + \beta\,\text{PS}_r(s_1) - \kappa\,k_r^{\text{innov}} - \phi\,k_r^{\text{enter}}, \tag{15}$$

which satisfies $\sum_r W_r = \text{CS}_{\text{total}} + \sum_r \text{PS}_r - \sum_r(\kappa\,k_r^{\text{innov}} + \phi\,k_r^{\text{enter}})$ by construction. The runner verifies this identity numerically and finds residuals at machine precision.

A.3 Evaluation by Monte Carlo with common random numbers

We average $K = 5000$ independent simulations per scenario. Each market draws its own initial state s_0 from the same distribution `simulate_data.jl` uses, so the counterfactual sits on the same population the model was estimated on. The same seed (20260424) drives both scenarios, and per-market we use `MersenneTwister(seed + k)`. Identical s_0 draws and identical EVT1 shock paths under both scenarios mean each market contributes a paired observation $(W_r^{\text{base}}, W_r^{\text{alli}})$, so the variance of the welfare difference is much smaller than the variance of either level. Headline numbers stabilise to about 0.2 percentage points by $K = 1000$.

A.4 Calibration and results

We hold the demand parameters, β, σ, ρ, c_o, $c_{n,0}$, and $N_{\max}$ at the values of Table 1, and use the structural estimates from Section 5: $\widehat{\kappa} = 0.2193$, $\widehat{\phi} = 0.1810$, and uniform $\widehat{\gamma}_r = 0.1500$.

Table 6 reports the full per-region decomposition. Innovation rises by +4.71 % in region 1 and +6.14 % in region 2, and falls by $|-3.89|$ % in region 3. Entry rates move in the same direction with smaller magnitudes.

Table 6: EU–US alliance: per-region equilibrium and welfare

	Region 1 (US)		Region 2 (EU)		Region 3 (RoW)	
Quantity	Baseline	Alliance	Baseline	Alliance	Baseline	Alliance
$P(\text{innov} \mid \text{old})$	0.3846	0.4027	0.3779	0.4011	0.3860	0.3710
$\Delta P/P_0$	+4.71%		+6.14%		-3.89%	
$P(\text{enter} \mid \text{pe})$	0.5544	0.5692	0.5766	0.5932	0.5757	0.5576
$\Delta P/P_0$	+2.67%		+2.89%		-3.15%	
PS_r	0.8500	0.9290	0.8471	0.9292	0.8787	0.8116
Costs paid	0.0820	0.0849	0.0861	0.0899	0.0867	0.0837
W_r	2.1481	2.2868	2.1411	2.2821	2.1721	2.1706
$\Delta W_r/W_{r,0}$	+6.46%		+6.59%		-0.07%	

Notes: Monte Carlo over $K = 5000$ independent markets, each drawing its own s_0 from the DGP used by `simulate_data.jl`; common random numbers across scenarios (seed = 20260424). Calibration: $\widehat{\kappa} = 0.2193$, $\widehat{\phi} = 0.1810$, uniform $\widehat{\gamma}_r = 0.1500$. Baseline blocs $(1, 2, 3)$; alliance blocs $(1, 1, 2)$. CS allocated equally across the three regions; PS_r aggregates own-region firms' profits in both periods; costs paid charges $\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}}$ to region r.

A.5 Discussion

Three points are worth highlighting.

Allied regions gain in two ways. Their pooled innovator count lowers $c_{n,1}$ and $c_{n,2}$ whenever *either* region has an innovator, raising the period-2 profit of being on new tech. This pulls more firms into innovation and entry there, which raises producer surplus, and the additional new-tech output also lowers prices globally, raising the regions' share of consumer surplus. Welfare rises by +6.46 % in region 1 and +6.59 % in region 2.

Region 3 loses on innovation but breaks even on welfare. Tougher global Cournot competition from cheaper allied competitors drags region 3's $P(\text{innov})$ and $P(\text{enter})$ down by about 4 % and 3 % respectively, and its PS_3 falls by roughly 7 %. Its share of the global CS gain is almost the same magnitude in the opposite direction, so its net welfare moves by less than a basis point (−0.07 %). The unallied region is harmed in real-economy terms (less innovation, lower profits) but the consumer-side benefit of cheaper goods produced by the alliance compensates almost exactly.

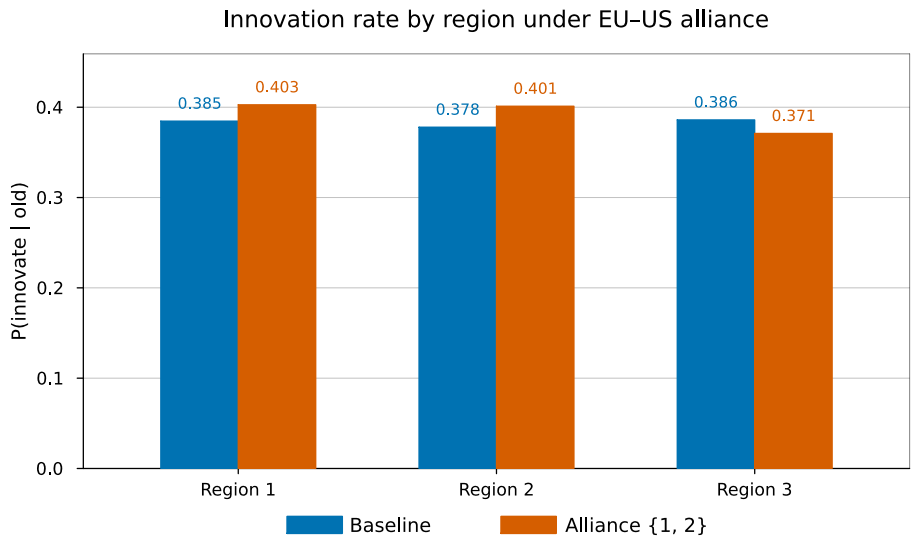

Figure 6: Innovation rate $P(\text{innov} \mid \text{old}, r)$ by region, baseline vs. alliance.

The aggregate gain is positive and sizeable. Total welfare rises by +4.31 % of baseline. Roughly half of the gain accrues to each allied region, with a near-zero residual from region 3. Knowledge integration is a positive-sum game even before factoring in the aggregate consumer benefits.

The result depends on the calibration. Under an earlier $(\gamma, \sigma) = (0.05, 1.0)$ setting the same alliance moved welfare by under 1 % everywhere; tripling γ scales the spillover delivered by the merger, and halving σ doubles the logit slope $P(1-P)/\sigma$ against any payoff change, so the combined effect on equilibrium quantities is roughly 6×. The recalibrated values stay well below the cost-floor saturation that sets in at $\gamma \gtrsim 0.30$, so the model is in its responsive interior. Full details, including the K-stability diagnostic and the $\sigma \cdot \gamma_{\text{em}}$ welfare convention, are in `notes/merger_counterfactual.md`.

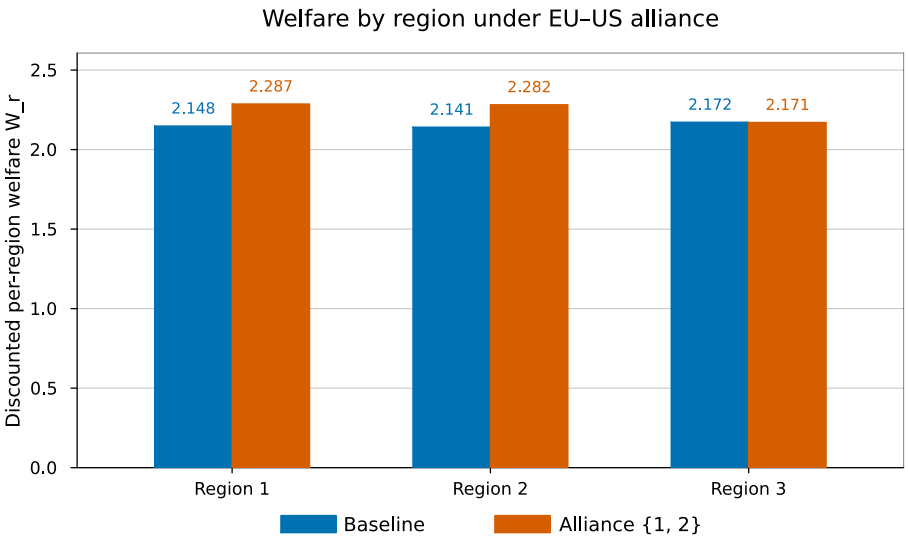

Figure 7: Per-region discounted welfare W_r baseline vs. alliance. Both allied regions gain visibly; region 3's bars are indistinguishable.